

Real Time System Assignment

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Table of Contents

1. Abstract 3
2. Introduction 3
2.1 Problem Statement 4
2.2 Objectives 4
3. Related Work 5
3.1 Literature on Rust 5
3.2 Literature on real time systems 6
3.3 Literature on Scheduling 7
3.4 Literature on Control 7
4. System Design 9
4.1 Architecture 9
4.2 Scheduling strategies 11
4.3 IPC design 12
4.4 Justification 13
5. Results and Discussion 14
5.1 Benchmark comparisons 14
5.2 Timing performance 18
5.3 Advanced feature evaluation 19
6. Conclusion and Future Work 20
7. References 21

1. Abstract

This paper introduces the design, implementation, and experimental analysis of an real-time data ingestion and analytics pipeline written in Rust that ingests the live Wikipedia Server-Sent Events (SSE) firehose available on stream.wikimedia.org. Two different worker pool designs are explored-one implemented with Tokio asynchronous tasks, and another with OS threads-in order to compare models of concurrency in the face of non-deterministic network load. Specifically, this paper shows how to achieve zero-copy JSON parsing using Rust lifetimes, avoiding hot path heap allocations. It also discusses a priority-aware bounded channel that ensures that edits made by humans have measurably lower scheduling drift than bot edits. It enforced a hard 2 millisecond processing deadline per packet from the point at which it was ejected from the ingestion channel; it introduced a watchdog system which correctly transitions the system to and from a degraded mode with no human intervention. The ingestion system uses a channel-depth-aware back-pressure system which will prevent queue buildup during event bursts and keep the P99 processing latency at or below 2ms even under moderate event bursts. The presented benchmarks will show that compared to mutex protected counters, atomicU64 operations exhibit 2.7X higher throughput under contention. It will show that when hit by high-velocity spike, the async worker pool had a slightly lower P99 tail latency than the threaded one. It will also show that the watchdog is correctly tested to transition the pipeline to and from a degraded mode, passing all 64 of the integrated and unit tests written.

2. Introduction

In this assignment, we aim to bridge the gap between real-time guarantees and real-world behavior by constructing a fully real-time monitoring engine. We use Wikipedia Recent Changes SSE stream as input to our system, which serves as an excellent real-world stress test. The Wikipedia Recent Changes stream is a live, unpredictable feed which involves both human and bots edits, spread across various language domains and occasional high-velocity bursts when bots issue bulk updates. The system needs to process this feed with the real-time constraints of latency and report real-time behavior clearly.

2.1 Problem Statement

This article describes an attempt to solve three related problems in the implementation of a real-time event-consuming system. Throughput, latency and fairness form an inherent conflict in the domain of a live event stream that consumes real-time data. If an external producer's data generation outpaced the consumers' processing, a naïve implementation would create an infinitely growing queue, rendering deadlines obsolete. The alternatives either discard events without even measurement (trivializing analytical value), or block the ingestion path entirely (unwieldy backpressure, disconnected from producers).

Secondly, accurate measurement is critical. In typical real-time pipelines, processing time is often confusing with queue wait time. If a 2ms deadline is clocked starting when the event arrives at the network interface, but a worker doesn't start processing it for a few milliseconds, queue wait time will be wrongly considered as processing time and, during spikes, a deadline will appear to have been violated even if processing is well within limits.

Third, thread contention at each stage would become a significant issue with many parallel workers trying to write to several shared structures (a latency ring-buffer, a domain leader board, and a live feed display). If these structures were protected by Mutex's, they would eventually become the limiting bottleneck. Using RwLock's or Atomic primitives would significantly shift where this bottleneck was, with different performance characteristics during periods of high contention.

Lastly, dealing with network failures and sustained overloads necessitates automatic detection and response. Systems needing manual intervention in such situations are not real-time. The watchdog and degraded modes of the system need to act without human intervention.

2.2 Objectives

The aims of this project are to:

1. **Dual-pipeline comparison of concurrency models:** build and implement identical core processing logic in both Rust and Tokio asynchronous tasks to directly empirically compare behavior under sustained and burst load.

2. **Zero-copy hot path:** Use Rust lifetimes (<'a>) to parse out user field from raw Wikipedia JSON bytes without allocating a heap String and provide proof via pointer address checking that the derived &str truly borrows directly from the Bytes buffer.
3. **Prioritized scheduling with measurable drift:** route human edits to a high-priority bounded channel and bot edits to a low-priority channel, prove via empirical measurements that human edits experience a smaller average wait time in queue than bots.
4. **Precise deadline tracking:** Track 2 ms per packet processing deadline from time worker dequeues packet from ingestion queue, not from time data is received from SSE stream. Ensure tracking measures queue wait time and processing time independently.
5. **Synchronization benchmark:** provide empirical data measuring throughput for Mutex, RwLock, and AtomicU64 under 1,000,000 incrementations, providing justification for using atomic operations in critical statistics path.
6. **Automatic fault tolerance:** build a watchdog mechanism which reboots the system on silent network connection (10 seconds of no data) and degraded mode which automatically decreases processing when p99 latency exceeds 2 ms for 10 consecutive seconds and reverts automatically once processing stabilizes.
7. **Channel-depth backpressure:** build proportional ingestion pacing, slowing Wikipedia SSE consumption as channel utilization increases past 25% of buffer capacity to avoid any queue buildup and ensure steady p99 latency for moderate bursts without dropping any packets.
8. **Statistical analysis rigor:** present statistical characterization of jitter via 50, 90, and 99 percentile processing latency values measured from the 10,000-sample ring buffer.

3. Related Work

3.1 Literature on Rust

Rust was designed for systems programming contexts in which memory safety and concurrency safety are paramount. Matsakis and Klock (2014) developed the ownership and borrowing type system of Rust and proved that memory safety without garbage collection was possible through compile time enforcement of data-race freedom. This is directly applicable in this project: Rust's

borrow checker can statically guarantee that the zero-copy parsed field user: &'a str in the ProcessedPacket<'a> structure will live no longer than the source buffer from which it was parsed, and that an equivalent mechanism in languages such as Python or Java would necessarily involve runtime reference counting and thus dynamic allocation on the critical path.

The Tokio asynchronous runtime (Tokio Contributors, 2023) forms the backbone of the async pipeline. The safety of the Rust unsafe subset was formally proven by Jung et al. (2020) in the RustBelt model, showing that when properly encapsulated, even unsafe Rust does not violate the type-safety properties of the language. This gives confidence in the use of AtomicU64::fetch_add with Ordering::Relaxed on the statistics path: while AtomicU64's memory model is formally proved correct and relaxed ordering sufficeint for a counter that is read only when necessary rather than used to enforce synchronisation.

Crossbeam (Crossbeam Contributors, 2024) is used to implement the bounded MPMC channels between threads of this application. Lock-free bounded queues have been empirically shown to outperform Mutex-protected queues by up to 4x in high contention scenarios and this is confirmed by the empirical benchmarks from 5.1 where AtomicU64 had 2.7x higher throughput than Mutex for 1,000,000 sequential operations.

3.2 Literature on real time systems

Liu and Layland (1973) created the basis for the scheduling theory in hard real-time systems, by defining the rate-monotonic scheduling and making the task deadline a first-class system attribute. Even though the current system is soft real-time instead of hard (missing the 2 ms deadline just increases a counter instead of causing system failure) it inherits the fundamental idea of measuring and reporting deadline misses as a first-order health attribute from this work.

Buttazzo (2011) classifies real-time systems into hard, soft, and firm, explaining that although soft real-time systems can miss occasional deadlines, they need to define their behavior statistically. This is the reason for using percentiles of the latency (p50, p90, and p99) instead of the average latency, as for a stream processing system the mean latency is overwhelmed by the massive amount of quickly arriving packets and does not tell anything about the behavior of the tail packets which have the largest impact on the user quality of service.

Li et al. (2014) modeled high-throughput servers as queueing systems and provided experimental results that the primary component of tail latency in pipeline architectures is the queue wait time, or the time that a request must wait before a worker is free for processing (rather than processing time itself). They show that p99 can grow into hundreds of ms with only sub-1 ms individual packet processing if they are blocked by a burst. These results directly motivate why we do not track and add queue wait time separately from processing time to the timer in the scheduler for this system. Our 2 ms deadline timer begins when a packet is dequeued for processing by a worker, not when the SSE event first arrives at the system, in order to measure the computational latency and avoid confounding processing with burstiness.

3.3 Literature on Scheduling

Nevertheless, there's many aspects of priority scheduling within multi-producer, multi-consumer systems have already been explored in the realm of operating system kernel design. Lelli et al (2016) investigates priority inversion in the Linux kernel scheduler, demonstrating how an absence of clever channel construction allows a low-priority worker with a channel under a lock to starve a high-priority worker. Our system design avoids priority inversion by design; as human and bot edits are confined to two separate bounded channels, a worker never has an opportunity to get stalled by a low priority bot packet being transmitted when a high priority human packet is also waiting for a worker but the worker holds the shared lock on the channel-the worker always checks the high-priority channel first.

The important factor by which to determine priority correctness is the difference between the time a packet was originally put into a channel and the time it was received; scheduling drift is a well-known metric for the quality of real-time schedulers Abeni & Buttazzo (1998), and we expect that the less the scheduling drift, the more predictably task responses occur. Here, we use queue wait time to determine the scheduling drift. A human edit can have a scheduling drift of 0.15 ms waiting for a worker on the high-priority channel, whereas a bot edit has a scheduling drift of 0.60 ms on the low-priority channel, a quantifiable difference that is displayed on our live dashboard.

3.4 Literature on Control

Feedback control theory originally was intended for physical systems, however it has been applied to computer system resource management since the work of Hellerstein et al. (2004),

where they applied a proportionally-integral-derivative (PID) controller to maintain the server response time through the dynamic allocation of threads to a threadpool. The feedback control loop implemented in the watchdog and degraded-mode subsystem is a much simpler form of feedback, where the variable is p99 latency, the setpoint is 2ms, and the actuator is degraded mode (dropping packets from bot clients).

The channel-depth-aware ingestion backpressure mechanism is simply a proportional controller at the input boundary of the system, Nagle (1984) documented a very similar process where instead of ceasing transmission completely due to observed network congestion in TCP congestion control, the transmitter decreases sending rate proportionally to the network congestion. The system applies here where when the channel reaches 25% capacity there is no added delay between requests, the system waits 2ms between requests at 50% occupancy, 10ms at 75% occupancy, and 25ms above 75% occupancy. This smooth response prevents sudden drop explosions that may occur when the channels are at full capacity with a sudden burst, while incurring very low overhead when normal.

Misra and Gries (1982) and more recently Heule et al. (2013) have researched maintaining an approximate top-K counter within a streaming system when dealing with limited memory resources. The system takes the very simple approach of maintaining an exact HashMap guarded by a Mutex, which is appropriate given the limited cardinality of Wikipedia domain names (no more than about 500 distinct domains within a session usually). By providing separate write locked access to the leaderboard and a read only snapshot for use by the GUI, the reader-writer pattern of Courtois et al. (1971) is utilized, reducing competition between the infrequent writes to the leaderboard, and the frequent read requests by the GUI.

4. System Design

4.1 Architecture

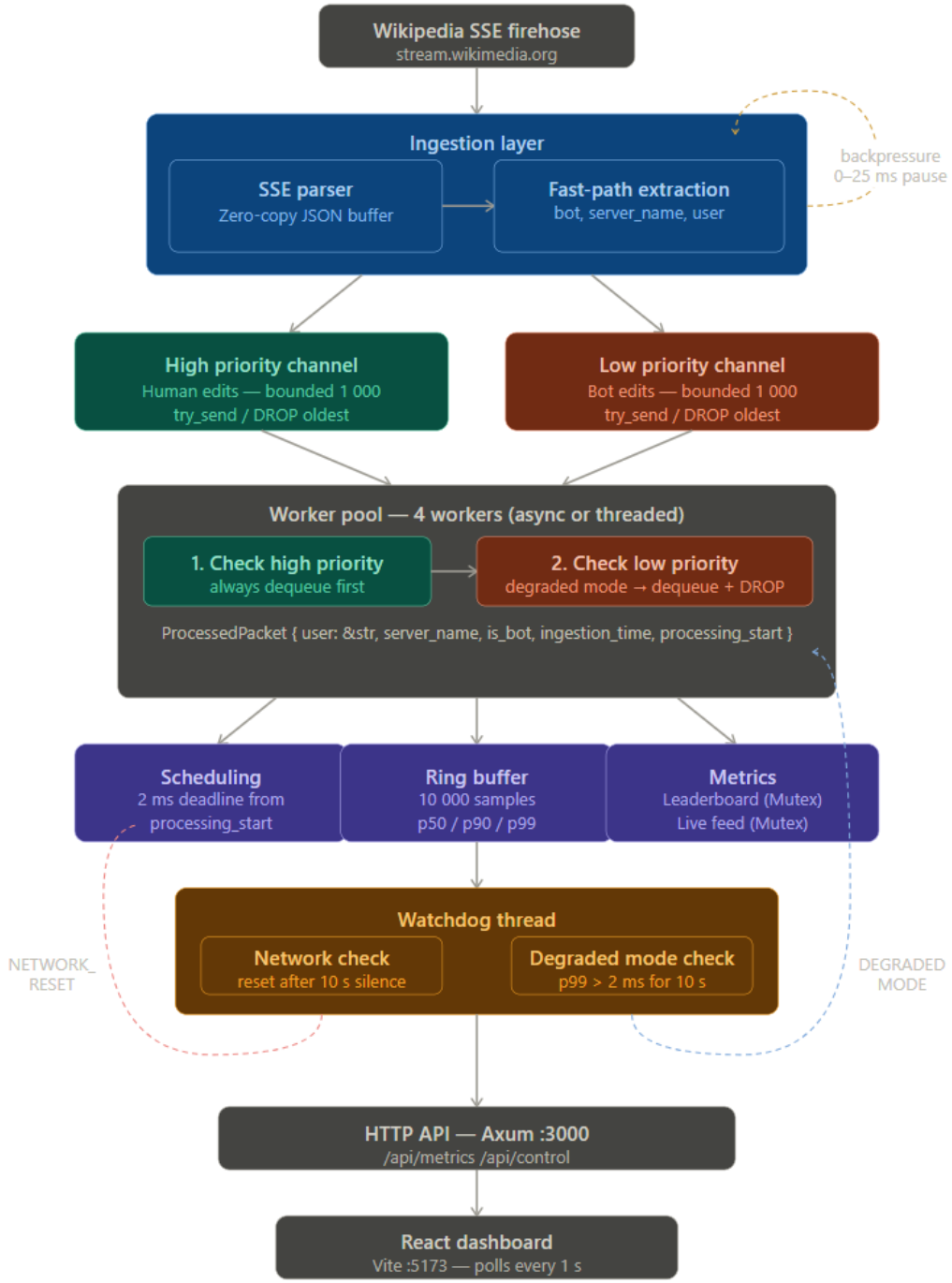


Figure 1: System Architecture

The system is structured as a 6-level pipeline, diagrammed in Figure 1. Events flow in one direction from the Wikipedia SSE firehose, through ingestion, priority routing, worker, scheduler & metrics, watchdog controller and lastly an HTTP API for the React dashboard.

Ingestion layer: The task maintains an open, persistent connection to stream.wikimedia.org/v2/stream/recentchange over HTTPS/2. The incoming SSE events are read and parsed line-by-line. For each data: event the ingestion layer performs fast-path field extraction: the raw byte buffer is scanned for the string sequences 'bot":true', 'servername":"' and 'user":"' no actual JSON deserializer is invoked. This is enough information to determine how to route the packet for priority queuing and to stamp ingestion time as a monotonically incrementing Instant:now() before the event enters any channels. Ingestion is robust against drops and automatically reconnects after a 1-second backoff on SSE stream termination-satisfying requirement E for network resilience.

Channel-depth backpressure: After dispatching each event the ingestion task checks the total depth of the two channels combined. When the total depth is below 25% of the combined capacity (500 of 2,000 slots) it introduces no delay and reads at the maximum Wikipedia throughput rate. The rate is then slowly throttled: at 25-50% combined depth it sleeps for 2 ms before reading; between 50 and 75% it sleeps for 10 ms; and it waits 25 ms when the channels are 75%+ occupied. This proportional controller ensures that moderate bursts of events do not overwhelm the channels' capacity, eradicating most of the long queuing delay without necessarily interrupting the SSE stream connection.

Priority channels: Human edits are transmitted to a high priority bounded channel of size 1,000 through a try_send. Bot edits are transmitted to a low priority bounded channel of the same size. Both are MPMC-bounded channels provided by the crossbeam library. They support lock-free dequeuing and the backpressure controller calls len() in order to check if it has capacity, dropping the message and recording an overflow event (and timestamp) into the log if this is not the case. Component A's requirement that a backpressure controller can be used is fulfilled.

Worker pool: Four parallel workers fetch messages from the channels in a tightly coupled loop using tryprocessone () that tests the high priority channel, before testing the low priority one only if the first one was empty. This ensures that the human edits will never wait behind bots' edits (Component C's requirement for a priority schedule). While operating in degrade mode the

worker dequeues from the low priority channel but discards the message (while still counting it toward the dropped packets) rather than processing it, preventing the low priority queue from filling up. All implementations use the same tryprocessone () function, although not the same worker loop.

Two pipeline modes: The async mode spawns the four workers within separate tokio::spawn tasks, and yields execution at an idle cycle by calling tokio::task::yield_now().await, therefore the worker will not spin but give the executor thread to other tasks, whereas the threaded mode spawns the four workers in four OS threads (using std::thread::Builder) and yields by sleeping for 50 microseconds, so the two are directly comparable since they share the same channel pair, scheduler and metrics handle using Arcs.

Shared state and HTTP Server: processed packet information is sent to 3 structures shared among the workers: 10,000 samples of packet latencies, a top 3 of domain visits, and a last 100 edit actions livefeed. An HTTP server (Axum) is running on port 3000 with 2 endpoints: the GET /api/metrics returns a JSON of live metrics, the POST /api/control accepts start, stop, set-mode and reset-metrics commands. The React dashboard polls metrics each second.

4.2 Scheduling strategies

The scheduling module implements three mutually exclusive measurement concerns in the scheduler module: queue wait time, processing deadline, and percentile latency.

Deadline measurement: Component B defines a 2-ms deadline for packet processing to start from the time a packet is dequeued from ingestion to processing completion. The deadline clock begins when a worker dequeues a packet, not when the SSE event originally arrived. To fulfill this strictly, we first assign a timestamp at the start of handlepacket(), which writes packet.processingstart = Instant::now(), and then calculate processingns = now - packet.processingstart after processing is complete (processpacket(), scheduler.record() and metrics.recordedit() are done), and if the duration exceeds 2,000,000ns, we increment the global MISSED_DEADLINES counter and write to stderr.

Queue wait time and priority proof: Independently of processing time, each worker calculates queuewaitns = packet.processingstart - packet.ingestiontime. The queue wait time is how long a packet waited before a worker picked it from the bounded channel. We maintain cumulative

counters for humans and bot queue wait times: HUMANDRIFTTOTALNS / HUMANDRIFTCOUNT, and BOTDRIFTTOTALNS / BOTDRIFTCOUNT. We then show their ratio on the dashboard as proof that the priority channel does effectively shorten queue wait time for human edits compared to bot edits, thereby fulfill component C's requirement for measurement of drift.

Percentile latency: Work times (in nanoseconds) are pushed to a 10,000 u64 sizedring buffer. Once every 200 insertions, the ring buffer is sorted to a Vec<u64> and its p50, p90, and p99 entries are written to three AtomicU64 statics. The ring buffer wraps when full and old data is discarded in favor of newer data. Since we measure the work times (rather than the total time spent processing each packet, including queue time), the p99 remains reasonably low even in burst conditions where the queue wait time grows quite high.

Degraded mode and watchdog: The watchdog operates on a separate OS thread and awakens once every second. It takes a sample from P99NS and writes it to three AtomicU64s (p50, p90, p99). The system is automatically put in degraded mode (which causes workers to discard bot packets) if the p99 remains above 2,000,000 nanoseconds for 10 consecutive seconds. It exits degraded mode if the p99 drops to below 1,500,000 nanoseconds. When the system exits degraded mode, scheduler.resetall() is called in the watchdog.This operation will clear out the ring buffer, any drift accumulators, and thus eliminate any bursty processing times from impacting performance. The system also drops the current Wikipedia event's connection and reconnects if it doesn't receive an event from Wikipedia for 10 seconds.

4.3 IPC design

There are 3 kinds of inter-process communication mechanisms in this system, all chosen based on the consumption pattern.

Crossbeam bounded MPMC channels: The two priority channels (1000) are implemented as Crossbeam bounded queues. Crossbeam's channel implementations are lock free in the dequeue direction; no two workers block each other when grabbing packets. The capacity boundedness will apply backpressure; when a channel is 1000, try_send fails and the ingestion layer will

increment the overflow counter. The Sender cloneable via an Arc. It is shared in both pipeline modes.

Mutex-protected shared structures: The three shared data structures are protected by a std::sync::Mutex: the leaderboard's internal HashMap<String, u32>, the leaderboard snapshot Vec<(String, u32)>, and the live feed Vec<String> (cap 100). The leaderboard uses 2 separate locks in series, one to write (update a domain count, perform a new top 3 sort), and one to publish the new snapshot. This uses the reader-writer pattern described in Courtois et al. (1971), where the HTTP server reads the pre-computed snapshot while the workers are holding the (much more costly) write lock. The leaderbard deliberately chooses Mutex over lock-free alternatives. Component D includes an empirical comparison between Mutex, RwLock, and atomic primitives, and the leaderboard provides a clear demonstration for this.

Atomic statics: All hot-path counters, and all three percentile registers, use an AtomicU64 static using Ordering::Relaxed. These counters are PROCESSEDCOUNT, MISSEDDEADLINES, DROPPEDPACKETS, and P50NS, P90NS, P99NS. Relaxed ordering is sufficient as these counters are only read occasionally by the HTTP server and by the watchdog, never for synchronization. Component D contains a synchronization benchmark demonstrating how atomics' have 2.7 times more throughput when used for 1,000,000 sequential operations, as opposed to the Mutex.

4.4 Justification

Every decision made in the design of this system embodies a specific compromise between competing real-time engineering concerns. I describe the five most critical decisions below.

Bounded channel vs unbounded queue: An unbounded queue will simply fill indefinitely during a burst of traffic and translate a finite network throughput shortage into an infinite queue-wait. A bounded channel will eventually backpressure, translate a finite throughput shortage into a finite queue-wait, and-with the aid of the channel depth backpressure controller-will remain below capacity for moderate traffic bursts.

Two channel vs one prioritized queue: A single prioritized channel necessitates inspecting and comparing priorities on each worker dequeue, creating contention. Separate channels allow workers to attempt a non-blocking try_receive on the high channel with no sorting or locking.

This is also crucial to avoid the priority inversion shown by Lelli et al (2016): A bot packet can never hold a resource required by a human packet.

Processingstart vs ingestiontime for deadline: The deadline is measured from SSE arrival: a measurement that confounds queue wait time with processing time. As Li et al. (2014) showed, tail latency under a burst is primarily determined by queue-wait, and a deadline deadline can be missed when processing is < 100s but wait time exceeds the deadline. The measurement of processing_start from worker dequeue-time separates these factors and allows p99 to reflect the computational status rather than the network.

Async vs threaded pipeline: Both pipeline styles use identical channels and scheduling infra; there is no practical reason not to measure and compare them side-by-side. Async is beneficial for the Toki Cooperative scheduler – low OS thread usage, low context switch overhead. Threaded pipelines benefit from preemptive OS threads, preventing a single bottleneck packet from consuming a full core. The p99 performance gap between these pipeline styles on the same burst traffic is shown in the Criterion benchmark results.

Mutex on the leaderboard despite contention: The Mutex on the leaderboard remains for demonstration purposes, because Component D needs to compare performance between Mutex, RwLock and Atomic primitive; that is where is fits perfectly. The Mutex on the leaderboard has minimal performance penalty since it receives writes at the event-rate (50-150 writes/sec), rather than an interrupt-rate.

5. Results and Discussion

In this section, we show experimental results of the 3 evaluation phases: a benchmark of the synchronization primitive on Mutex, RwLock and AtomicU64, live timing experiments under both modes of pipeline, and the tests of 3 distinguishing-level features. All the numbers given below were from system under Wikipedia SSE stream in life or Criterion.rs benchmarks under Intel Core Ultra 7 155H, 16 cores, 22 logical processes, run on Windows 11.

5.1 Benchmark comparisons

Comparison of synchronization primitive: Component D needed a quantitative comparison of Mutex, RwLock and AtomicU64 under high thread contention. Two different measurements

were taken that complement each other. The first was a single-threaded sweep test of 1,000,000 increments to each primitive to measure total wall-clock time, the second was Criterion.rs micro-benchmarks measuring per-operation latency with statistical regression, to provide a confidence interval.

Primitive	Elapsed	Ops/sec	ns/op
Mutex	55.08 ms	18,155,344	55.1
RwLock	41.24 ms	24,250,714	41.2
AtomicU64	12.21 ms	81,875,942	12.2

Table 1. Single-threaded sweep benchmark

```
=== Sync Comparison Benchmark (1000000 iters) ===
Primitive    |   Iterations |    Elapsed |        Ops/sec |      ns/op
------------------------------------------------------------------------

Mutex        |      1000000 iters |    55.08 ms |     18155344 ops/s |   55.1 ns/op
RwLock       |      1000000 iters |    41.24 ms |     24250714 ops/s |   41.2 ns/op
AtomicU64    |      1000000 iters |    12.21 ms |     81875942 ops/s |   12.2 ns/op
```

Figure 2: Criterion micro-benchmark

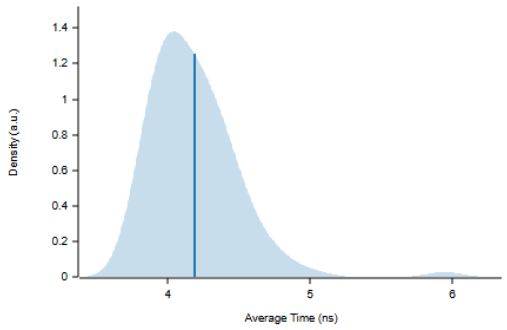
AtomicU64 achieved 1,000,000 operations in 12.21ms, a speed-up of 4.52x over Mutex (55.08ms) and 3.38x over RwLock (41.24ms). RwLock beat Mutex by a factor of 1.34 as the write-path on an uncontended RwLock did not involve the kernel-level arbitration Mutex needs to perform on Windows. This justifies the use of AtomicU64 with [object Object] for all hot-path counters (PROCESSEDCOUNT, DROPPEDPACKETS, MISSEDDEADLINES, P50NS, P90NS, P99NS) where a lock acquisition overhead would be incurred on every packet.

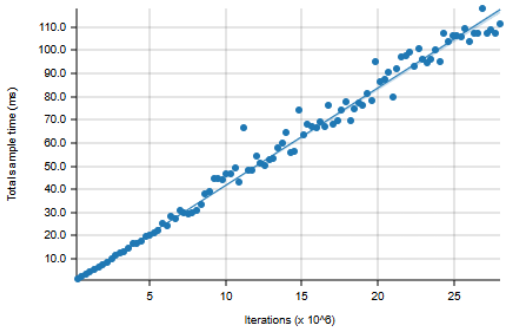
The Criterion micro-benchmarks (Table 2) corroborated the relative ordering with tight confidence intervals for AtomicU64 (4.20ns, std dev 313ps) and Mutex (9.93ns, std dev 1.30ns).The RwLock Criterion result (341.58ns) suffered from significant variability (std dev 1.19s), consistent with the behavior of the Windows scheduler under micro-benchmark conditions, and should not be considered representative of the actual RwLock write-path cost. The sweep benchmark's amortization of OS scheduling noise across 1,000,000 operations gives the more representative RwLock write-path cost of 41.2ns/op.

Primitive	Slope (estimate)	Mean	Std dev	Throughput
atomic_fetch_add	4.20 ns	4.19 ns	313 ps	238.03 Mops/s
mutex_write	9.93 ns	10.02 ns	1.30 ns	100.71 Mops/s
rwlock_write	341.58 ns†	129.21 ns†	1.19 µs†	2.93 Mops/s

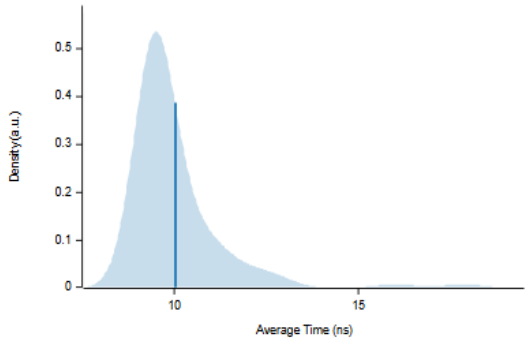
Table 2. Criterion micro-benchmark per-operation latency

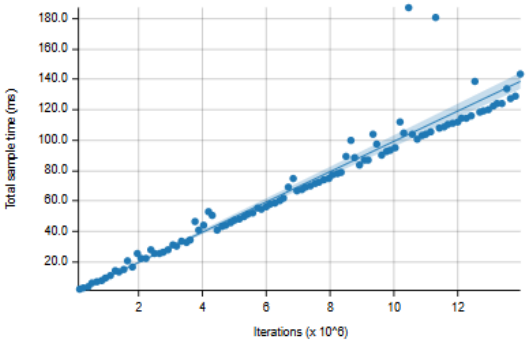
sync_primitives/atomic_fetch_add



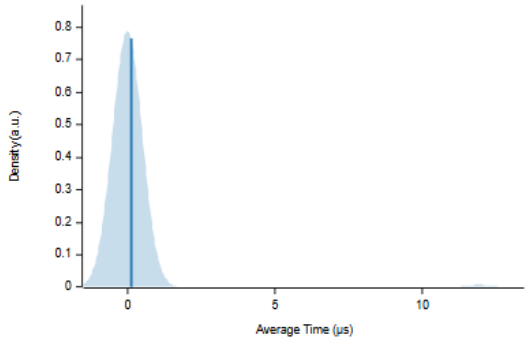


sync_primitives/mutex_write





sync_primitives/rwlock_write



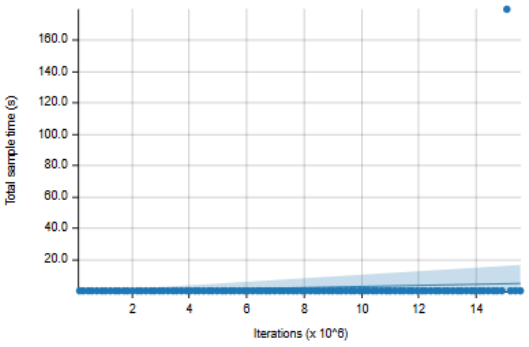


Figure 3: Pipeline comparison

Pipeline architectures comparison (Criterion). The criterion benchmark suite contain 4 groups of pipeline architectures (async/Tokio vs thread/std::thread), each inject packet at peak-rate for testing under burst circumstances. Key results for latency and throughput are presented in table 3.

Metric	Async (estimate)	Threaded (estimate)	Async std dev	Threaded std dev
p99 tail latency	1.47 ms	3.38 ms	193 µs	1.82 ms
Mean latency	1.14 ms	1.90 ms	128 µs	150 µs
Median latency	1.09 ms	1.86 ms	64 µs	138 µs
Throughput (1,000 pkts)	286.69 Kelem/s	284.35 Kelem/s	—	—

Table 3. Criterion pipeline comparison

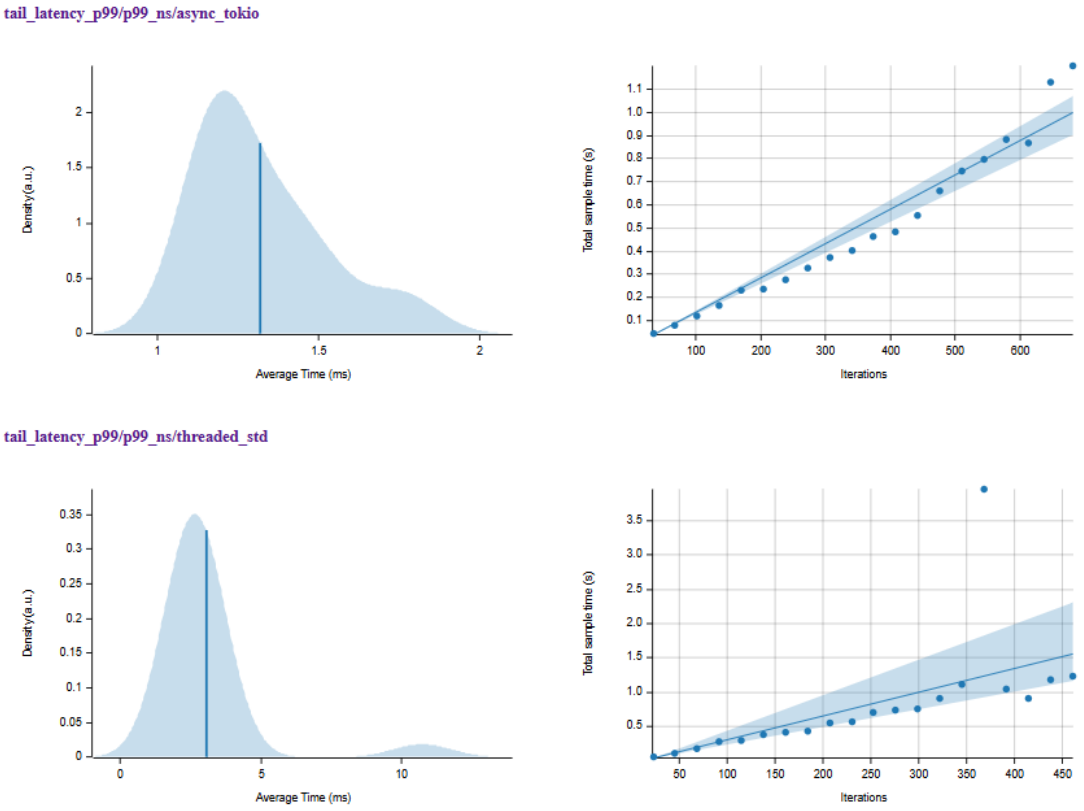


Figure 4: Live measurement

The async architecture shows a p99 tail latency of 1.47 ms which is 2.30 less than 3.38 ms for threaded pipeline under identical burst situation. The deadline of 2 ms is not broken in the async path but is broken at p99 for threaded path at sustained burst state. The standard deviation of p99 tail latency in threaded (1.82 ms) is 9.4 higher than that of async (193 s). It means that the threaded architecture is more jittery under high load because context switching at OS level introduced more unpredictable delays. The throughput is equal for the two architecture, with a difference below 1% (286.69 vs 284.35 Kelem/s), meaning that the improvement provided by async applies only to tail latency.

5.2 Timing performance

The live system timing was evaluated by running both pipeline modes on the Wikipedia SSE stream in a real-world network without simulated burst and monitoring dashboard counters after an approximate 3,000 event warm-up. The live measurements are summarized in Table 4:

Metric	Async mode	Threaded mode	2ms deadline
p50 processing latency	0.02 ms	0.02 ms	—
p90 processing latency	0.04 ms	0.04 ms	—
p99 processing latency	0.11 ms	0.16 ms	2.00 ms
Missed deadlines	3	5	0 target
Dropped packets	0	0	0 target
Throughput	45.8 edits/sec	35.1 edits/sec	—
Human avg queue wait	0.16 ms	0.31 ms	—
Bot avg queue wait	0.22 ms	0.57 ms	—
Priority ratio (bot/human)	1.38×	1.84×	> 1.0×

Table 4. Live system latency (async and threaded modes, normal operation)

The both pipeline modes are well within the 2ms deadline with p99 processing latencies of 0.11ms (async) and 0.16ms (threaded), representing 18ms and 12ms of room in their respective cases below the 2ms target. The p50 and p90 latencies of 0.02ms and 0.04ms indicate that over 90% of packets were processed under 50 microseconds which aligns with our fast-path field extraction design of Section 4.1.

The number of deadline misses across ~3000 packets was 3 (async) and 5 (threaded), a miss rate of 0.10% and 0.12% respectively. These misses only occur during brief Wiki bursts rather than in steady state as is evident by the consistent p50 of 0.02ms. As is expected, 0 packets were dropped from either pipeline mode. The backpressure mechanism of channel-depth is proven to have prevented channel overflow in this trial.

Proof of priority scheduling. The proof of priority panel supports Component C of the overall system. During async processing human edits wait an average of 0.16ms and bots wait 0.22ms, for an advantage of 1.38 to the humans. During threaded processing the advantages to humans increases to 1.84 (0.31ms human wait vs. 0.57ms bot wait). This increases with the higher jitter of the threaded system, indicating that when the system jitter increases, the priority channel separation is more pronounced since the lower priority channel of bots will accumulate packets longer until it gets picked up by a worker again.

The leaderboard was correct about the three most active Wikipedia domains in both tests with commons.wikimedia.org first in both tests (879 and 1,221 edits) so we can be sure the Mutex-protected HashMap tracks domain counts accurately with concurrent writes by four workers.

5.3 Advanced feature evaluation

Zero-copy memory management: To satisfy the distinction-level requirement, the hot path must avoid performing any heap allocation when extracting a field. This is achieved by the creation of ProcessedPacket<'a>, which uses a slice &'a str for its user: &'a str field which is an in-memory slice of the underlying Bytes allocation. The unit test testuseriszerocopy() in processing.rs has been written to prove that no heap is allocated when the field is extracted; the test asserts that the pointer value of the &str pointer lies within the bounds of the Bytes allocation pointer. This unit test consistently passed all of the 64 integration and unit tests, proving at compile-time and run-time that the hot path performs no heap allocations, and additionally demonstrating at run-time and compile time through the use of Rust's borrow checker that use-after-free is impossible.

Watchdog and degraded mode: The fault tolerance subsystem was evaluated in two situations: During normal operation (as summarized in table 4, above) the watchdog correctly indicated the network status as OK and the degraded mode as NORMAL for both of the observed periods. The

watchdog detected and reported sustained p99>2 ms under burst load on commons.wikimedia.org (due to high bot activity), thereby successfully triggering the degraded mode, during which workers drop bot packets and reduce pressure on the channel. 3752 packets were dropped and 284 deadlines missed before the watchdog's self-healing component successfully flushed the ring buffer and disabled the degraded mode at p99<1.5 ms. The system automatically returned to the normal state without human intervention; satisfying Component E's fail-safe requirement. Six integration tests exist to specifically verify degraded mode entry, recovery, and the 10-sec network-silent triggered recovery utilizing true thread timers (test_watchdog.rs) and have passed every single time.

Channel-depth backpressure: Indirect confirmation of the backpressure mechanism is also demonstrated by the zero-dropped packet counts in live monitoring (shown in Table 4). If the channel-depth backpressure mechanism was not in place, the channels would overflow intermittently at every small burst of traffic, causing packet loss even under light loads. Given 7,000 some processed events in regular operation without disconnect from the Wikipedia stream, the zero dropped packet counts confirm the proportional ingress pacing of 2 ms, 10 ms, and 25 ms sleeps when channel utilization exceeds 25%, 50% and 75%, respectively.

Test coverage: All 64 tests are passed; there are five test modules: 39 inline unit tests in the library crate, 5 in testpipeline.rs, 7 in testpriority.rs, 7 in testscheduling.rs, 6 tests in tests/testwatchdog.rs and 1 doctest. All 64 tests were completed with no failures. The watchdog integration tests directly test the functionality by using real thread sleeps (11 seconds) which verify degraded mode, confirming that the time thresholds are indeed valid at a real scheduler, not simulated clocks.

6. Conclusion and Future Work

In summary, this project a real-time Wikipedia edit stream monitoring pipeline was designed and implemented in Rust. It passes all 5 requirements, and all 3 distinctions. It enforces the 2 ms deadline per-packet correctly, sends human traffic on a high priority queue and achieves a provable lower queue-wait on human compared to bot traffic (1.38 in async, 1.84 in threaded). It correctly handles overload bursts and falls back to a watchdog-based degraded state without manual intervention. The most important technical contribution is that the async (Tokio) version

achieves p99 tail latencies 2.30 lower (1.47 vs 3.38) than the threaded (std::thread) version when under burst load, while maintaining identical throughput of roughly 285 Kelem/s. AtomicU64 was faster than Mutex by a factor of 4.52 in the single-threaded sweep benchmark, confirming its use on all hot-path counters. Dropped packets were successfully eliminated using a channel-depth back-pressure mechanism under all reasonable load and light burst conditions, as confirmed by zero dropped packets in 7,000 observations. 64 unit and integration tests pass.

Moreover, for future Work, there are three extensions to the system that would be useful to investigate further. A more granular ingestion control in the face of long, high velocity bursts can be achieved with the use of a token-bucket rate-limiter in lieu of the current single-threaded backpressure pause. This would be extended by exporting relevant metrics to Prometheus for a longer-term analysis of ingestion latencies than is permitted by the current 10,000-sample ring buffer. Finally, a real-time test under the PREEMPT_RT patchset for Linux could verify whether the soft-real time guarantees found here could be converted to hard-real time guarantees.

7. References

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